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AI-Driven Root Cause Analysis for Well Failure Events

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Abstract

Predicting well failures and identifying their root causes are critical for maintaining efficient oil and gas operations. This research explores the use of an AI-driven framework for root cause analysis, aiming to detect potential well failure events, explain their causes, and support proactive prevention. The framework leverages operational data such as well events, issue classifications, engineers' notes, and applies a large language model to analyze when, where, and why failures might occur. By training on historical data with optimized hyperparameters, the framework demonstrated consistent improvements across evaluation metrics. Evaluation metrics including lexical and human evaluation metrics featured the high performance of Llama 3.3 70B model highlighting the benefits of prompt engineering, fine-tuning, combining structured and unstructured data. The approach proposed reduces manual effort, improves safety, enables cost savings, and supports data-driven maintenance planning. Field engineers validated the framework's effectiveness, highlighting its practical value. Overall, the research contributes to the advancement of well failure analysis, and demonstrates its practical applicability and reliability in real-world scenarios. Ultimately, the implementation of our AI-driven framework has the potential to transform well operations, enabling data-driven decision-making and improved asset performance.

Keywords: Well Integrity, Failure Identification, Root Cause Analysis, Large Language Model (LLM)

Introduction

Well failure refers to the failure or collapse of barriers in a well designed to control the flow of fluids, resulting in the release of oil, gas, or other substances into the environment involuntarily [1]. In the oil and gas industry, such failures present significant operational and environmental risk, including contamination of groundwater resources, uncontrolled migration of fluids, and potential safety risks to operators. These failures are either due to leakages, mechanical issues such as improper cementing, poor wellbore geometry, chemical degradation like casing corrosion and failure of sealing materials, and even biological processes, such as damage caused by some types of bacteria [2]. With the severity and complexity of well integrity failures, sound capability for proper detection, and analysis of well failures is more critical for ensuring long-term well performance, protection of the environment, and overall safety of oil and gas operations.

Overview on the paper structure

The paper is organized as follows. Next section provides a literature review of some relevant research papers that delves into various debates and key concepts related to well integrity management and the application of Large Language Models (LLMs) in anomaly detection. This is followed by the methodology section, which outlines the experiment plan, dataset description, model design, and failure analysis generation. The results and evaluation section discusses the experimental setup, training and validation, result analysis, model evaluation, time cost reduction, and field operator feedback. A human evaluation is also included to assess practical usability. Finally, the paper concludes by summarizing the findings and suggesting future research directions.

Problem Definition & Objectives

Problem Definition

Well failures in oil and gas operations pose serious risks to safety, the environment, and overall system performance. These failures often result from mechanical faults, chemical deterioration, or biological influences, and they can lead to the unplanned release of oil, gas, or other hazardous substances. Such incidents not only endanger personnel and ecosystems but also disrupt operations and increase costs.

Accurately identifying and anticipating these failures remains a major challenge. Traditional root cause analysis and fault detection methods often struggle with limited precision, scalability, and responsiveness. One key obstacle lies in the complexity of the operational data itself, which includes both structured formats (like well event logs and failure codes) and unstructured content (such as engineers' field notes). This makes it difficult to extract timely insights using conventional tools.

In response to these limitations, this study explores a data-driven approach to predicting and analyzing well failures. We introduce a framework built on the LLaMA 3.3 70B model developed by Meta. This large language model is fine-tuned using domain-specific prompts and historical well data, including logs with deep temporal relationships and technical terminology relevant to field operations.

Our results show that the proposed model achieves improved predictive accuracy when compared to baseline methods. It also performs well in identifying underlying causes of failures, as confirmed through validation with experienced field operators. This approach supports more informed maintenance decisions and helps operators detect critical failure patterns earlier, contributing to safer and more reliable oil and gas production.

Objectives

The primary objectives of this research are to:

- Develop an AI-Based Root Cause Analysis Framework: Design and implement a framework that leverages a large language model (LLM) for analyzing structured and unstructured operational data to detect well failure patterns and identify root causes.
- Improve Failure Prediction Accuracy: Train and fine-tune the model using historical well data and engineers' notes to enhance the model's prediction accuracy and reduce false positives in failure detection.
- Support Data-Driven Maintenance Planning: Enable proactive maintenance strategies by providing actionable insights that help field engineers make informed decisions, ultimately reducing downtime and operational risk.

- Validate Practical Applicability: Conduct field validation to assess the model's real-world performance in supporting safety, cost savings, and decision-making effectiveness in oil and gas operations

Literature Review

Traditional methods for detecting and analyzing well failures have several limitations, including high costs, measurement errors, and incompatibility with non-piggable pipelines. In recent years, the application of artificial intelligence (AI) and machine learning (ML) techniques has shown promising results in improving the accuracy and efficiency of well failure analysis. By examining the existing body of research and identifying areas for further investigation, this literature review seeks to contribute to the advancement of well integrity management and the development of more effective and efficient methods for predicting and analyzing well failures.

Well Integrity

Well integrity is the ability of oil and gas wells to isolate zones and limit unintended fluid migration for the lifecycle of the well [1]. It is important for safety and environmental protection, particularly for aging and offshore assets. According to [3], approximately 30-45% of active wells are experiencing integrity issues, with the primary concern being sustained annulus pressure (SAP), which indicates a continuous leak path from the reservoir, highlighting a significant challenge in maintaining well integrity. [4] further identifies critical challenges impacting well integrity including issues with well head components, downhole safety valves, gas lift inspection valves and other suspension related activities. In addition to mechanical failures, cement sheath degradation, casing corrosion, and micro-annuli formation are major factors contributing to well integrity loss. [5]. For example, in the 2010 Deepwater Horizon oil spell, poor cementing practices led to loss of zonal isolation and a catastrophic well failure [6]. Failures in well integrity can also have extensive ramifications beyond operational concerns. As noted by [7] and [8],, such failures may endanger human life, cause significant environmental harm, result in damage to critical assets and equipment, and erosion of corporate reputation.

To address these challenges in a systematic way, well integrity is managed proactively through the process referred to as Well Integrity Management (WIM). This system is defined and implemented across all phases of the well life cycle from the basis of design and construction, to operation and intervention, and finally abandonment to ensure integrity during every phase [9]. Each of these phases follows specific technical requirements and verification processes which were all directed towards creating and maintaining effective barriers to protect against unintended fluid migration while ensuring operational safety and environmental protection. WIM also serves broader objectives within the oil and gas sector. It both reduces the potential risk of uncontrolled hydrocarbons and enables the lifespan of a well to be extended, creating maximum reservoir recovery and economic value [10]. Ultimately, the integrity of the well through its life provides the base for safe and reliable production operations to occur. Therefore, WIM will include technical, operational, and strategic aspects that advance safety to people and the environment and improve long-term productivity from oil and gas assets. However, while WIM is effective, there is still significant room for improvement, especially in terms of early identification of anomalies and potential integrity threats. This is the point where the implementation of advanced machine learning technologies can be transformative and enhance well integrity management.

The use of LLMs For Anomaly Detection

Large Language Models (LLMs) are a subset of Natural Language Processing (NLP) tools, characterized by their ability to comprehend and generate coherent, context-specific text, enabling the production of intuitive and human-like responses [11, 12]. Additionally, LLMs are able to recognize and analyze complex patterns and trends in large datasets, anticipate potential anomalies, and facilitate informed decision-making [11].

According to [13], there have been notable breakthroughs in LLMs, particularly GPT and Llama models, which have demonstrated substantial enhancements in text understanding and generation tasks. Notably, Llama distinguishes itself through its ability to process complex data formats in real-time, achieving this with remarkably low computational requirements and reduced resource utilization [11]. This capability enables Llama to perform proactive risk assessment, leveraging historical data to identify potential risks, ultimately enhancing predictive accuracy [11].

Numerous studies, including those by [11, 12, 14, 15], have investigated the application of artificial intelligence in predicting failures based on historical data and identified trends. For instance, research by Benard Ongwae [11] demonstrated the potential of tailoring LLMs to streamline risk evaluation processes, enhancing predictive analysis and enabling the development of preemptive strategies. The findings indicate that using LLMs increases the efficiency of incident classification [11]. Furthermore, the study revealed a significant reduction in analysis time, with tasks that previously required five working days to complete manually being accomplished in a remarkably shorter timeframe of about an hour with AI assistance [11]. In line with these results, Fahim Sufi's [12]'s research on classifying cyber incidents have also demonstrated the effectiveness of LLMs in generating comprehensive insights and identifying related features from textual data with high accuracy rates. The application of LLMs has been shown to facilitate the extraction of meaningful insights from textual data, enabling the accurate classification of incidents and prediction of underlying causes.

In conclusion, the literature review highlights the importance of well integrity management in the oil and gas industry, with a focus on the application of Large Language Models (LLMs) in anomaly detection and root cause analysis. The review demonstrates that LLMs have the potential to improve the accuracy and efficiency of well failure analysis, enabling proactive maintenance and reducing the risk of well failures.

Methodology

This section provides a detailed description of the experimental setup, dataset, and model architecture used to analyze well failure events and identify root causes, laying the foundation for the results and discussion that follow.

Dataset Collection

For the purposes of this study, we leveraged a real-world dataset consisting of operational data from around 1000 wells within the Middle East. The dataset contains over 80,000 records of operational data for these wells during the period of approximately 2 years. Key components of recorded data include well activities (i.e., well work logs), operational status, failure types, lost time incidents and a detailed description of the activities performed prior to and during the failure events. This data was collected by field operators throughout the regular monitoring of the well and for reporting purposes. Due to confidentiality agreements, the exact source details cannot be disclosed.

Notably, the dataset contains a wide variety of failure scenarios, the complexity of operations in the field can produce various kinds of well failure patterns for the operation of a well. It is the wide variety of operations that provide a strong opportunity to identify patterns, trends and contributing factors leading to well failures in different contexts.

Because of the nature of the analysis and the reliance on a heavily text-based unstructured data, we conducted an evaluation of the coherence and quality of narrative entries. A data validation process was conducted to check for internal and structural consistency for cross entries. This was a form of multi-dimensional quality review under which completeness (ensuring all relevant fields were filled), coherence (ensuring that narratives were in coherence with timestamps, failure categories, and other relevant operations data), and consistency (standardized use of the record terminology and categorizations across entries) were checked. These steps were important in protecting the integrity of the dataset and assuring precise, as well as, insightful analysis.

Below, in Table 1, shows the list of key fields in the dataset, along with their descriptions:

Table 1—Description of Dataset Fields for Well Operation Job Records

Field Name	Description
Site	A numerical identifier for the site where the well operation is taking place.
W_GNR_NAME	The name of the well or the general name of the operation being performed.
JOB_OBJ	A brief description of the objective or purpose of the job being performed.
JOB_SUB_CATGY	A sub-category or specific type of job being performed, such as "Demobilizing" or "Rigging Up".
RPT_DATE	The date on which the report or operation was performed.
JOB_SMRY	A brief summary or description of the job being performed.
JOB_DUR	The duration of the job being performed, in hours and minutes.
JOB_ENTR_STATUS	The status of the job entry, such as "OP" (operational) or "LT" (lost time).
DESCRIPTION	A detailed description of the activities performed during the job.
Old_Segment	An older or previous segment or category related to the job.
Segments	The current segment or category related to the job.
Year_Date	The year of the operation.
C_Duration	The duration of the operation in a specific format (e.g., 6:00:00 AM).
OP Hrs	The number of operational hours worked during the job.
DT Hrs	The number of downtime hours experienced during the job.
LT Hrs	The number of lost time hours due to accidents or other incidents.
NPT Hrs	The number of non-productive time hours experienced during the job.
segmentRank	A ranking or prioritization of the segment or category related to the job.
TopSegment	The top or most critical segment or category related to the job.
Segment	The current segment or category related to the job.
Start of the Week	The start date of the week during which the operation took place.
End of Week	The end date of the week during which the operation took place.
Updated_SOJ	An updated or revised version of the stage of the job.
OP per Month	The number of operational hours worked per month.

Data Preprocessing

A number of data transformation and preparation procedures were carried out in order to provide insightful analysis of well failure events. Records from several wells operating under various situations were included in the raw dataset; each well's data covered both regular operation times and episodes of lost time. We started by determining the lost time sequences for each well in order to gain a better understanding of each failure event. These sequences refer to times when there were interruptions or downtime in the well. To provide more context around each failure, we then included the most recent operational records that took place before each lost time. These records reflect periods when the well was functioning as expected. This allowed us to examine not only the failure but also the events leading up to it, which can provide important information about underlying causes and reoccurring trends.

After constructing these enriched sequences consisting of the failure period and its preceding normal operational records, we developed a method to extract and combine the corresponding job summaries. These summaries contain textual descriptions of the activities and observations recorded by field operators during both normal operation and lost time. By merging these entries into a single narrative for each sequence, we were able to capture the full context surrounding the failure event. Before using these summaries and feeding it into our LLM model, we performed an extensive text cleaning process. This included removing special characters, HTML tags, and other formatting issues that could interfere with data quality. This ensured that the text data was clear, consistent, and ready for subsequent analysis. An overview of the full preprocessing workflow is illustrated in figure [1].

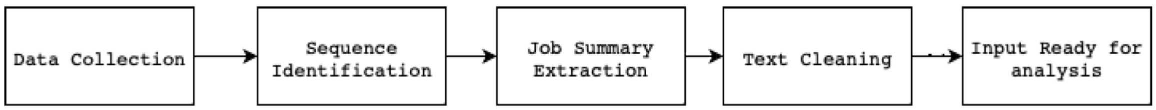

Figure 1—Data Transformation Pipeline for Well Failure Analysis

Model Design and Architecture

For this study, we utilized Meta's LLaMA 3.3 70B, a state-of-the-art decoder-only transformer model designed for high-performance natural language understanding. LLaMA 3.3 70B adopts a standard transformer architecture with key enhancements over its predecessor, including a 128K-token vocabulary for more efficient language encoding and grouped-query attention (GQA) to improve inference scalability. The model was pretrained on over 15 trillion tokens from high-quality, publicly available sources, including a substantial proportion of technical and code-related content. It was trained with a sequence length of 8,192 tokens and equipped with semantic deduplication and text-quality filtering pipelines, resulting in a cleaner and more contextually rich language representation. Given the complex and context-dependent nature of operational logs in well failure scenarios, LLaMA 3's extended context window and deep pretraining allow it to capture subtle operational patterns, temporal dependencies, and domain-specific terminology making it well-suited for our task of analysing well failures from unstructured data. These capabilities are summarized in Table 1, which outlines the model's key features, and their specific relevance to the problem of well failure analysis.

Table 2—LLaMA 3.3 Model Features and Their Relevance to the Study

Model Feature	Description	Application to the Problem
Expanded Vocabulary (128K tokens)	Recognizes a wider range of technical terms and rare tokens.	Enhances understanding of specialized oil and gas terminology in well operation logs.
Improved Attention Mechanism (Grouped-Query Attention)	Optimized attention mechanism for faster and more efficient inference.	Enables scalable analysis of large datasets with minimal performance bottlenecks.
Extensive Pretraining (15+ trillion tokens)	Trained on diverse, high-quality sources, including technical documentation.	Improves accuracy when interpreting complex, domain-specific operational language.
Long-Context Processing (8,192-token window)	Can analyze longer text segments in a single pass.	Captures dependencies across multi-step operational narratives and time-series events.
Clean and Deduplicated Training Data	Reduces noise and repetition in outputs.	Provides more accurate and contextually relevant failure analysis.

Failure Analysis Generation and Prompt Engineering

In the failure analysis generation stage, we utilized a Large Language Model (LLM) to produce structured explanations for each well failure event. Following data cleaning and preprocessing, we gathered previous summaries of the operational jobs preceding each failure, along with the failure's own text summary, in order to provide historical context. These inputs were dynamically integrated into a consistent prompt template designed to simulate a diagnostic setting, where the model assumed the role of a well operations engineer reviewing recent job activity to assess the root cause of a failure.

To ensure effective and structured failure analysis, the prompt was crafted to encourage expert reasoning, guiding the model to interpret symptoms, relate them to operational history, and propose a credible failure diagnosis. It was designed to utilize special system and user tags based on LLaMA's formatting conventions, helping to differentiate the roles for the system and the user. This approach enables the model to operate with a clearly defined role and task behavior, generating tailored responses based on dynamic user statements specific to each failure case. For clarity and consistency, the output was required to follow a strict structure comprising two elements: a single failure category drawn from a predefined list and an description beginning with a concise description of the event. The description also had to specify why the failure occurred, where it took place, and when it happened relative to prior operations. This structured format made the output more actionable for field engineers and aligned it with operational reporting needs.

Result and Evaluation

This section presents the outcomes of the experiments conducted to assess the performance of the proposed AI-driven root cause analysis framework, including the LLM evaluation metrics, as well as the results of the human evaluation and comparison with baseline models.

Experimental Setup

All of the experiments in this study were carried out in a Cloudera environment, making use of its strong infrastructure to test and validate machine learning models. Python 3.9 was used for the implementation, and Windows was used as the operating system. The computational resources utilized for this study consisted of 13th Gen Intel(R) Core(TM) i7-1365U vPro processor, NVIDIA GPU and a 32GB of RAM. A collection of specialized libraries, such as Pandas, Seaborn, and NumPy, were used for data analysis and manipulation as well as performance assessment. This setup and configurations enabled the efficient training and testing of the AI-driven root cause analysis framework.

Training and Validation

To create a strong basis for failure analysis, we devised a feedback loop and used it to refine the performance of the model, as shown in Figure [2]. The first part of the training involved engineering the prompt to get the LLM to be more accurate and relevant in its responses. The format of the prompt was designed to be similar to that used by industry-wide field operators when identifying and analyzing well failure events in addition to using language and terminology specific to the field. Next, we tested the outputs by performing cross-reference validation with the model responses compared to the field operator provided analysis, as this represented a well-established benchmark process for an assessment of the model. By comparing the model and manual generated analysis, we were able to assess the accuracy of the model in addition to identifying areas where it excelled and areas requiring improvement. Field operators also contributed to the model's validation by reviewing and providing feedback on the model's outputs using their expertise. This human-in-the-loop method ensured that the model outputs were precise and relevant in context. By continuously conducting this loop of feedback, we were able to enhance the model quality by refining the prompt which in turn resulted in a very functional framework that can provide accurate insights into the root causes of well failure events.

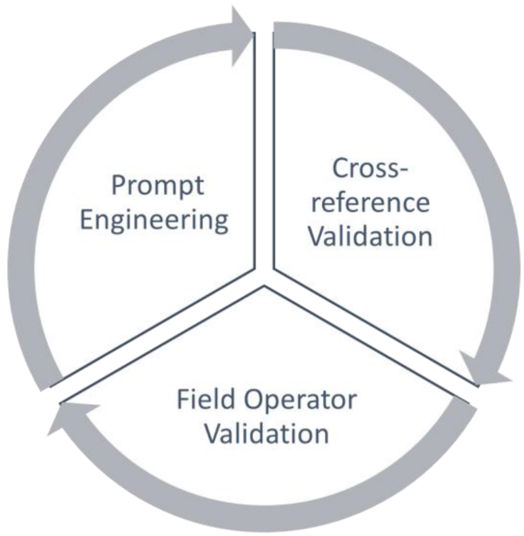

Figure 2—Robust Failure Analysis Framework

Analysis of the Results

Model Evaluation. To evaluate effectiveness and relevance of the model's outputs, a comparative analysis has been conducted between Llama 3.3 70B model and Mistral 3.1 24B model against the root cause analysis provided by field operators. 50 failure job summaries with human analysis were selected from the dataset for testing and validation acting as an evaluation benchmark. On average, a field operator takes about 15 to 20 minutes per case, which adds up to nearly 17 hours of manual work for the full set. In contrast, the models processed the same data in under 10 minutes total, highlighting a major advantage in terms of speed.

A mix of lexical evaluation metrics were used to compare the AI-generated outputs against the human-written analysis as outlined below:

BLEU Score. The BLEU metric ranges from 0 to 1, it measures how many words in machine-generated text appear in the reference human-generated text. It evaluates precision by counting the number of matching n-grams (combinations of n words) between the generated and reference texts. BLEU primarily focuses on precision but includes a brevity penalty to discourage overly short outputs that might otherwise be favored.

Key Components of BLEU:

- N-gram Precision: BLEU checks how often n-grams (word combinations) from the generated text match those in the reference text. This is usually done for n-grams ranging from 1 to 4 words.
- Brevity Penalty: A penalty is applied to short generated sentences that match well with the reference to prevent artificially high scores.
- Weighted Average: BLEU combines the precision scores for different n-gram sizes into a single score.

Formula for BLEU Score:

$$\boldsymbol{BLEU} = \underbrace{\boldsymbol{min}\left(\mathbf{1}, \boldsymbol{exp}\left(\mathbf{1} - \frac{\boldsymbol{reference-length}}{\boldsymbol{output-length}}\right)\right)}_{\textbf{Brevity penalty}} \underbrace{\left(\prod_{i=1}^{4} \boldsymbol{precision}_i\right)^{1/4}}_{\text{n-gram overlap}}$$

The formula has two components:

- **Brevity Penalty:** Penalizes translations that are too short compared to the reference, addressing BLEU's lack of a recall term.

- **N-Gram Overlap:** Measures how many word sequences in the generated text match those in the reference, serving as BLEU's precision metric [16].

ROUGE Score. The ROUGE score is primarily based on recall and was designed for text summarization, where the model-generated text is typically shorter than the reference text. ROUGE compares n-grams, word pairs, and word sequences between the reference and candidate summaries [17].

Key ROUGE Metrics:

- ROUGE-1: It measures the unigram overlap between the generated text and reference text. It is a straightforward metric that counts the number of matching single words.
- ROUGE-2: It measures the bigram overlap between the generated text and reference text. It counts the number of matching pairs of consecutive words.
- ROUGE-L: It takes the longest common subsequences (LCS), which are useful for capturing structural similarity between the generated and reference texts.
- Formula for ROUGE-N:

ROUGE-N Formula:

$$ROUGE-N = \frac{Number\ of\ matching\ n-grams}{Total\ n-grams\ in\ the\ reference}$$

ROUGE-L Formula (Longest Common Subsequence-based):

$$ROUGE-L = F_\beta = \frac{(1+\beta^2)\cdot P \cdot R}{\beta^2 \cdot P + R}$$

Where:

- P is precision
- R is recall
- is typically set to 1
- The range is from 0 to 1, where 0 indicates no overlap between the generated and reference text, and 1 indicates a perfect match.

Table 3 summarizes the average scores for both models:

Table 3—Lexical Evaluation Metrics Comparing Mistral 3.1 24B and Llama 3.3 70B Outputs

Metric	Mistral 3.1 24B	Llama 3.3 70B
BLEU	0.002	0.003
ROUGE-L	0.260	0.272
ROUGE-2	0.192	0.200

Although the lexical overlap with human-written text was generally low due to differences in expression, as seen in the BLEU scores, the ROUGE scorers indicate that both models are still able to capture parts of the intended meaning. Between the two, Llama 3.3 70B achieves slightly higher across all metrics compared to Mistral 3.1 24B. These small but consistent gains suggest that Llama 3.3 70B produces outputs that are marginally closer in wording and structure to the reference text. Nevertheless, the overall low lexical scores highlight that both models rely more on semantic alignment rather that exact word overlap to capture the essence of the root cause analysis

Human Evaluation. To further assess the quality of the generated root cause analysis, a human evaluation was performed on a sample of 50 well failure incidents. Each case included a sequence of previous operations, a summary of the failed job, and root cause outputs from the Llama 3.3 70B model.

The results show that the model performs well, with the evaluators praising the model's ability to provide robust explanations with only slight imperfections or areas for refinement. The evaluators' comments offer additional insight into the model's performance, with some noting that the explanations are "too lengthy" or "repetitive", while others appreciate the level of detail provided. Despite the minor flaws, majority of the model’s analysis were widely praised, with evaluators highlighting the model's ability to identify and explain complex failure causes. This suggests that the model is capable of generating high-quality root cause analyses that are satisfactory or better in most cases.

Overall, the human evaluation results demonstrate the model's potential to provide valuable insights into well failure incidents, helping to improve the efficiency and effectiveness of operations. With continued development and refinement, the model can become an even more reliable tool for root cause analysis in the oil and gas industry.

Conclusion and Future Direction

This research proposes a new AI-centric framework for root cause analysis of well failures in oil and gas industry. The framework uses LLAM 3.3 70B model to analyze operational data relating to well events which showed an improved capability for accurate failure predictions. The model achieves a high accuracy score after iterative prompt engineering, fine-tuning and field operators’ validation. This capability permits the identification of failure patterns rapidly and accurately enabling proactive, data-driven decisions to make wells safer and more efficient. This research advances well failure analysis and proves its practical application and reliability in oil and gas operations. Future research directions include investigating the applicability of the framework to other areas in oil and gas operations, including drilling operations and pipeline integrity monitoring, where root cause analysis of failures is critical to operational efficiency. By pursuing these future research directions, the AI-driven framework for root cause analysis of well failure events can be further enhanced, leading to increased efficiency, safety, and reliability in oil and gas operations.

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